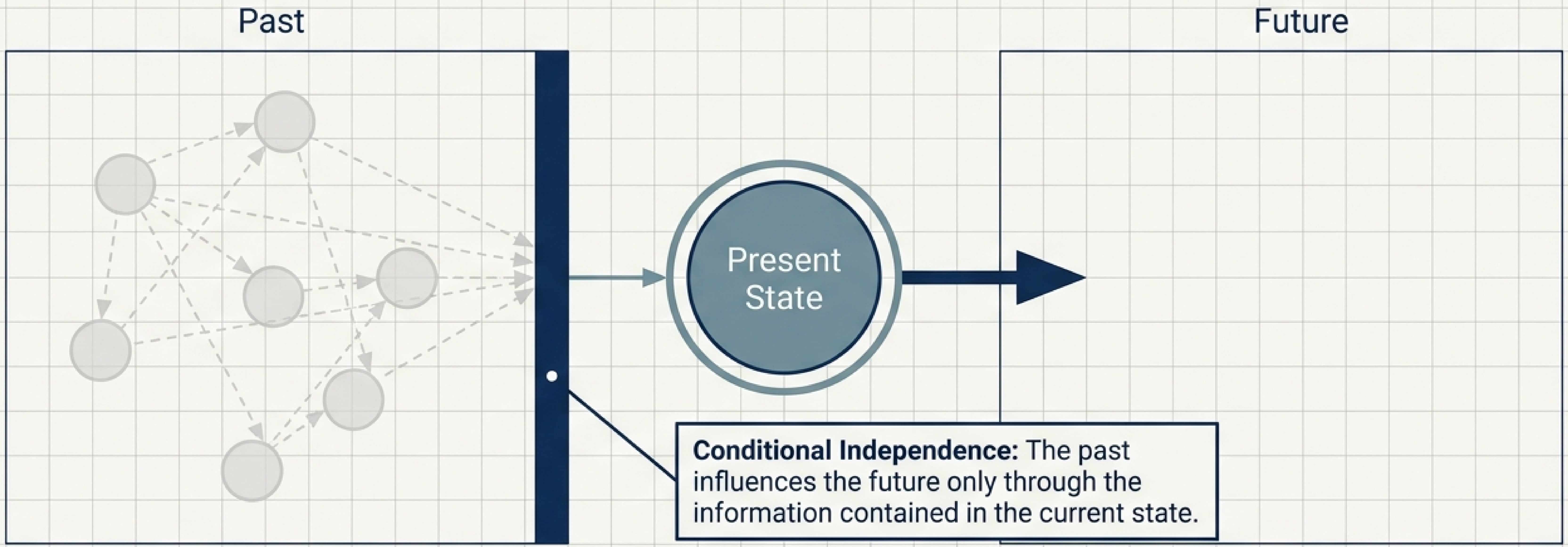
Translating Network Chaos into Clarity

A Diagnostic Framework Using Markov Chains

SYS_DIAGNOSTIC // CH. 3 // MARKOV_MODELS

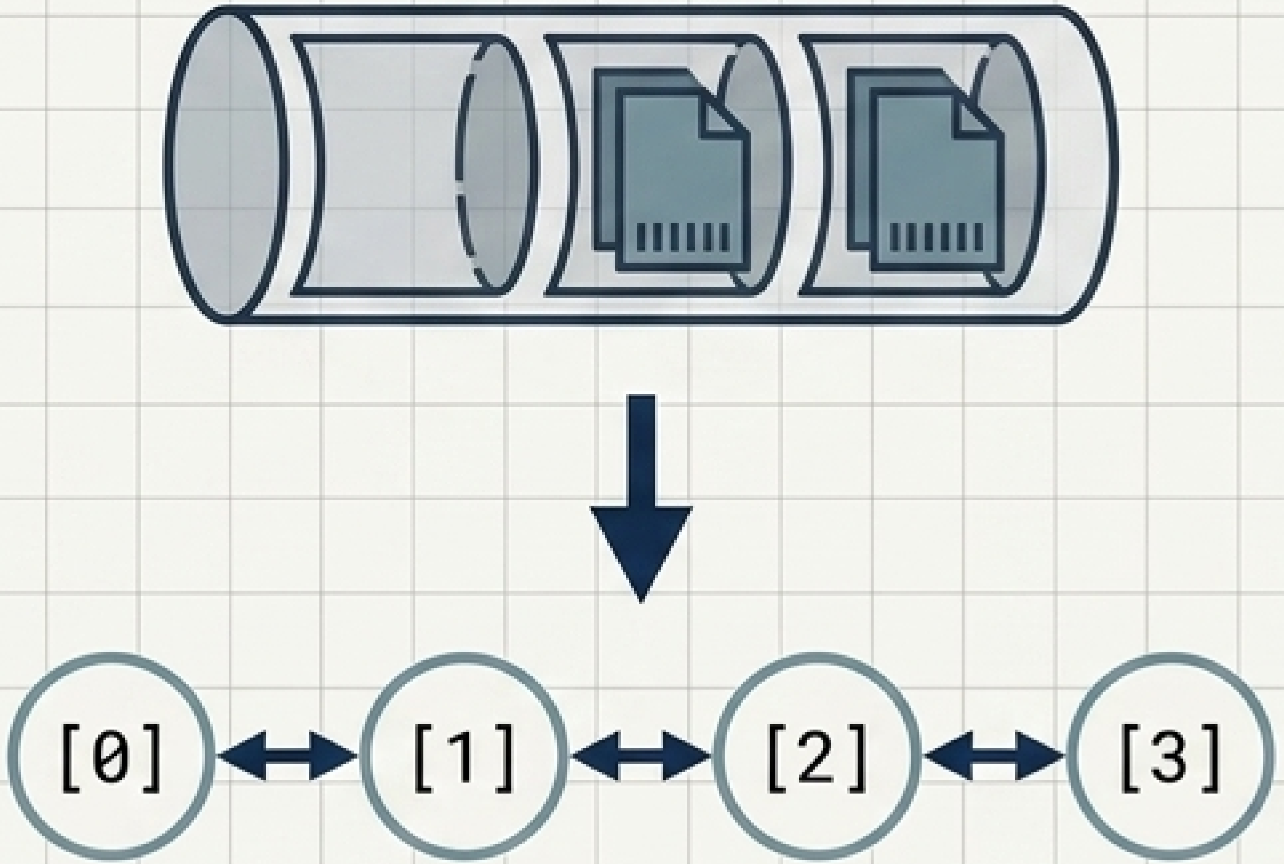
The Markov Property is a Time Filter



$$P(X_{n+1}=j \mid X_n=i,\ X_{n-1},\ \ldots,\ X_0) = P(X_{n+1}=j \mid X_n=i)$$

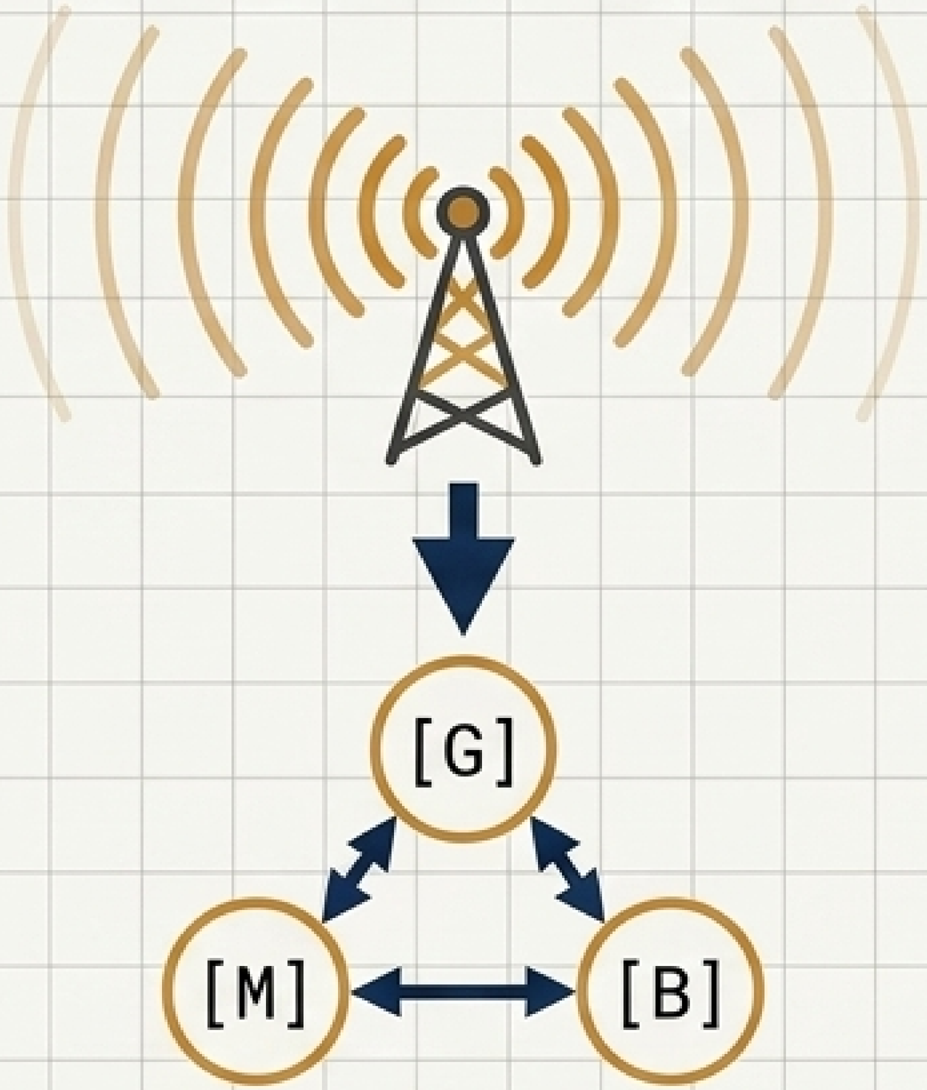
Defining the State Space (S)

Network Reality: Queue Occupancy



S = {0, 1, 2, 3} (Empty to Full)

Network Reality: Wireless Channel

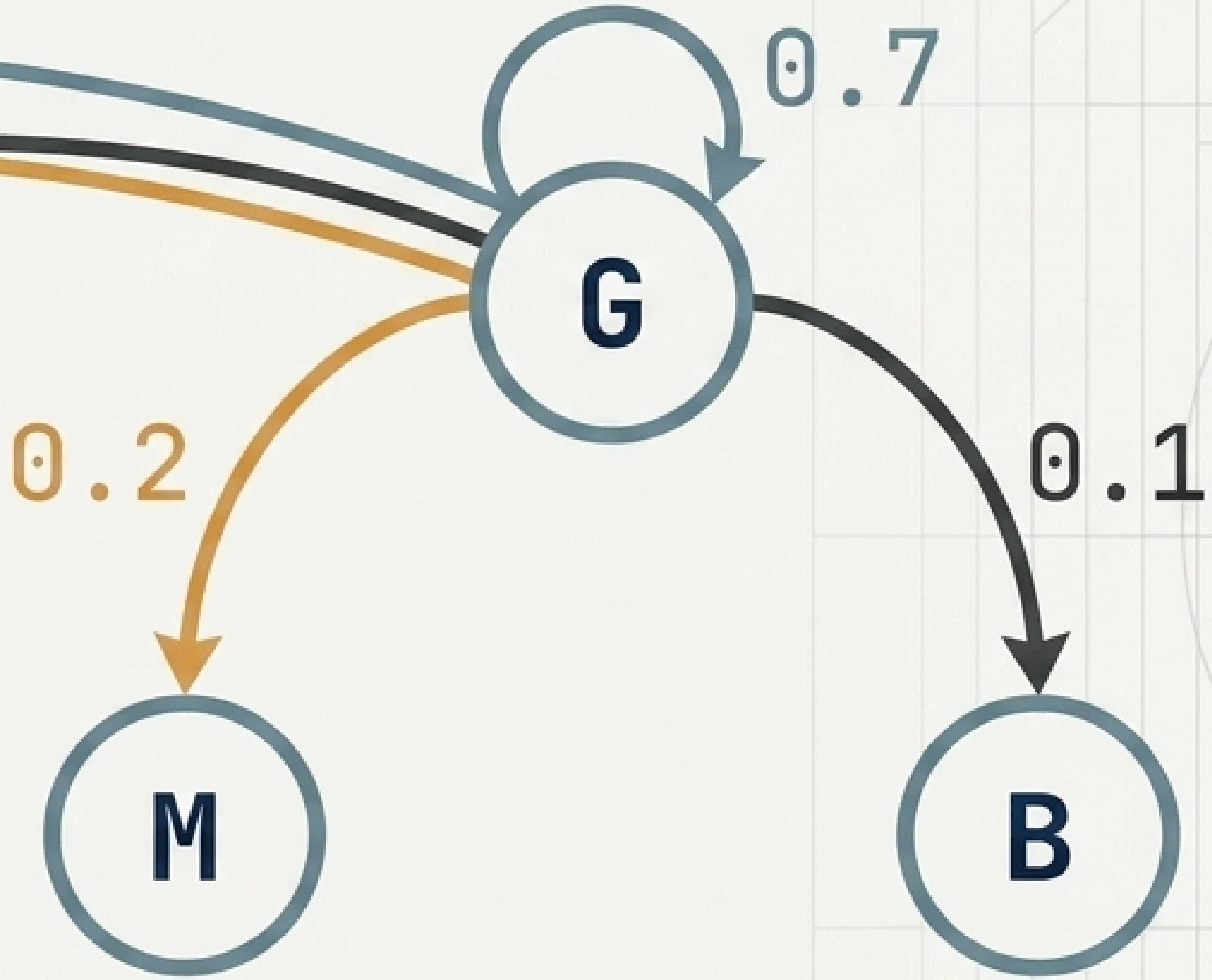


S = {G, M, B} (Good, Medium, Bad)

The state definition is the most critical modeling decision. A poorly selected state fails the Markov property by not capturing enough historical information to accurately predict the future.

Translating Matrices into State Machines

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$$



The Golden Rule of P: Every row must sum to exactly 1. Given state i, the system must transition to one of the possible states. $\sum P_{ij} = 1$.

The Great Divide: Discrete vs. Continuous

	DTMC (Discrete Time)	CTMC (Continuous Time)
Time Index	Discrete time instants	Arbitrary continuous times
Main Parameter	Transition Probability ($\mathbf{P}_{ij}$)	Instantaneous Rate ($\mathbf{q}_{ij}$)
Governing Matrix	Transition Matrix (**P**)	Infinitesimal Generator (**Q**)
Holding Time	Fixed step durations	Exponentially distributed
Ideal Application	Slotted systems, discrete frames	Asynchronous packet arrivals, failures

[DTMC -> P]

[CTMC -> Q]

Anatomy of Discrete Evolution (DTMC)

The Future State Distribution

The Current State Vector

The Transition Probability Matrix

$$\pi^{n+1} = \pi^{n)} P$$

1 Step: $\pi^{(1)} = \pi^{(0)} P$ → n Steps: $\pi^{(n)} = \pi^{(0)} P^{n}$

Chapman-Kolmogorov Note: Long-term transitions are constructed from shorter ones: $P^{(n+m)} = P^{n} P^{m}$.

Anatomy of Continuous Evolution (CTMC)

$$\mathbf{Q} = \begin{bmatrix} \boxed{\begin{matrix} q_{00} \\ q_{10} \end{matrix}} & \begin{matrix} q_{01} \\ q_{11} \end{matrix} \end{bmatrix}$$

Instantaneous Transition Rate (units: s^-1)

The Generator Diagonal. Must equal the negative sum of all outgoing rates: $q_{ii} = -\sum q_{ij}$. Every row sums to 0.

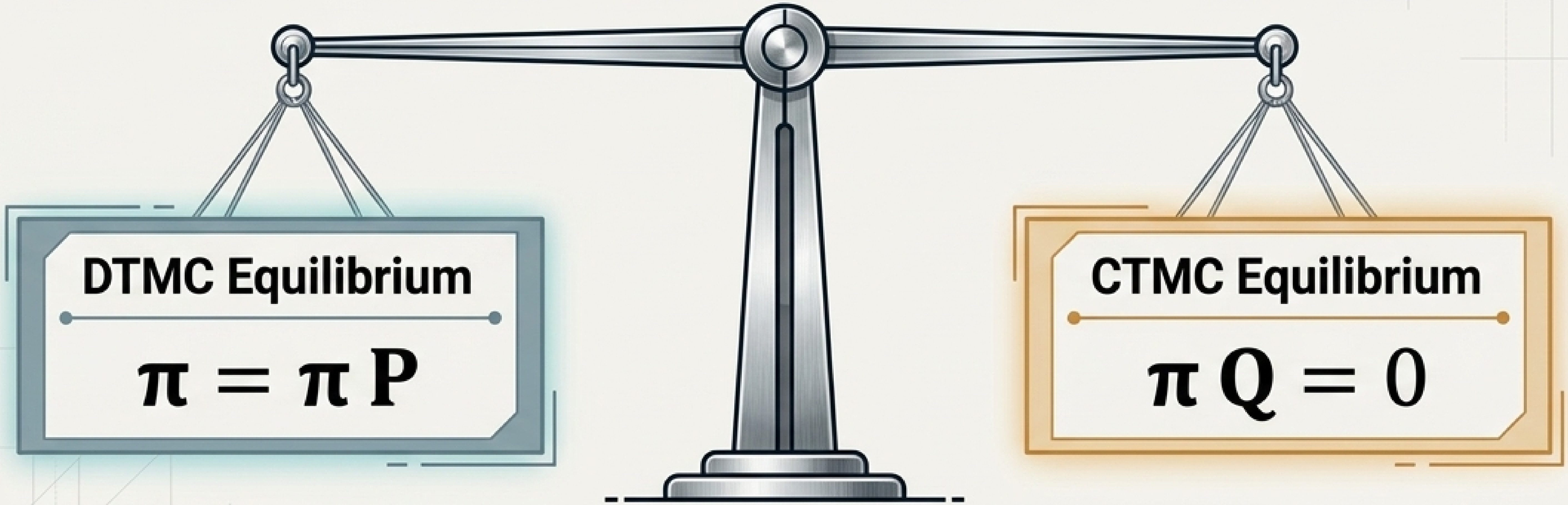
The Continuous Equivalent of Evolution

State Evolution: $\dfrac{d\pi(t)}{dt} = \pi(t)\mathbf{Q}$ → Solution: $\pi(t) = \pi(0)e^{Qt}$

(using the matrix exponential)

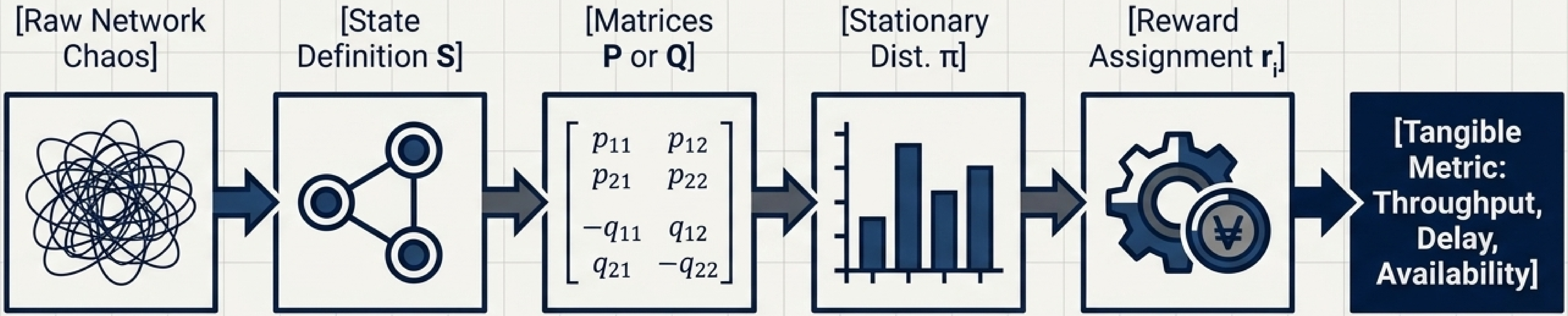
Finding Equilibrium: Stationary Distributions

Both branches represent the long-term fraction of time the process spends in each state.



Critical Constraint: Regardless of discrete or continuous time, the state probabilities must always normalize: $\sum \boldsymbol{\pi}_i = \mathbf{1}$ and $\boldsymbol{\pi}_i \geq \mathbf{0}$.

The Markov Reward Model Pipeline



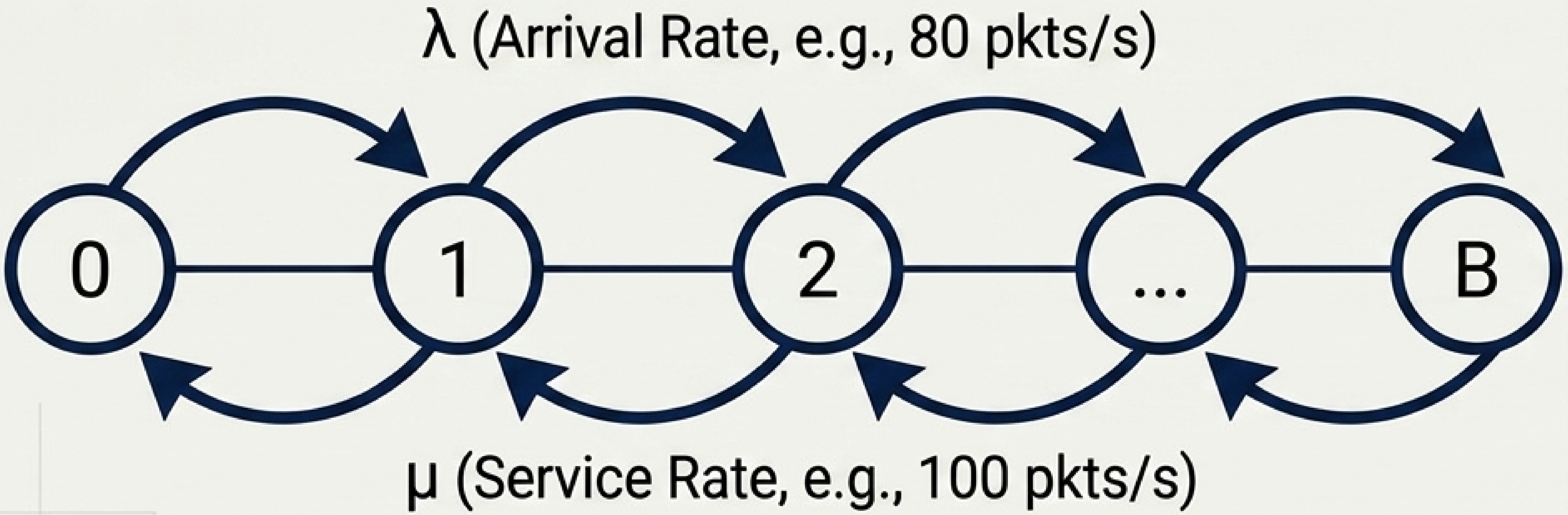
The Engine of the Pipeline

$R = \sum (\pi_i * r_i)$ → Average Reward = Sum of (State Probability × State Cost/Reward)

Translating Scenarios into States

Network Problem	State Definition (S)	Transition Triggers	Derived Metric
Buffer Queueing (M/M/1)	Number of packets (0, 1, ..., B)	Packet arrivals & departures	Delay, Packet Loss
Wireless Channels	Channel Quality (G, M, B)	Fading, interference	Avg. Throughput
Link Reliability	Link Status (Up, Down)	Hardware failures & repairs	System Availability
User Mobility	Connection (Conn, H/O, Disc)	Cell tower boundaries	Handover failure rate

The Birth-Death Balancing Act: M/M/1 Queues



Confinitrated **rewave** nenew nates, e.g., Pkts.

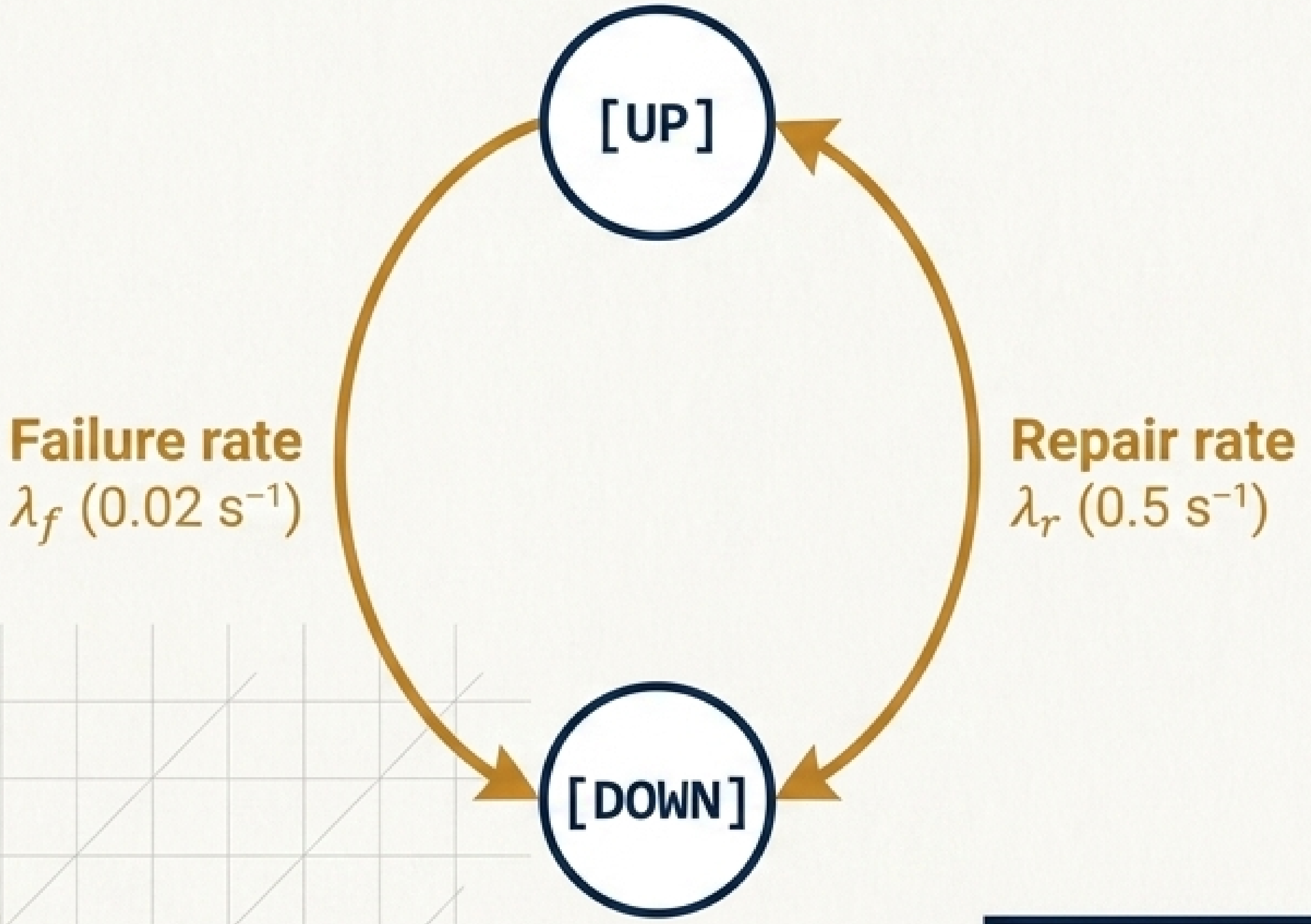
Finite Capacity limit. $P_{loss} = \pi_B$. Any arrival finding a full system is dropped.

Math Callouts

Stability Ratio: $\rho = \lambda / \mu$ (Must be < 1)

Occupancy Probability: $\pi_n = (1-\rho)\rho^n$ (Geometric distribution)

The Unavailability Seesaw: Link Reliability



CTMC Reliability Equations

Availability (A): $\left[\dfrac{\lambda_r}{\lambda_f + \lambda_r}\right] = \pi_U$

Unavailability (U): $\left[\dfrac{\lambda_f}{\lambda_f + \lambda_r}\right] = \pi_D$

$$\boxed{A + U = 1}$$

A powerful diagnostic tool for evaluating service level agreements.

Calculating Average Throughput (DTMC Reward)

Part 1

Stationary Probabilities (π)
Node G (0.6) | Node M (0.3) | Node B (0.1)

State Rewards (R)
Node G (20 Mbit/s) | Node M (10 Mbit/s) | Node B (2 Mbit/s)

Part 2

The Calculation

$$E[R] = \pi_G R_G + \pi_M R_M + \pi_B R_B$$
$$E[R] = (0.6 \times 20) + (0.3 \times 10) + (0.1 \times 2)$$
$$E[R] = 12 + 3 + 0.2$$

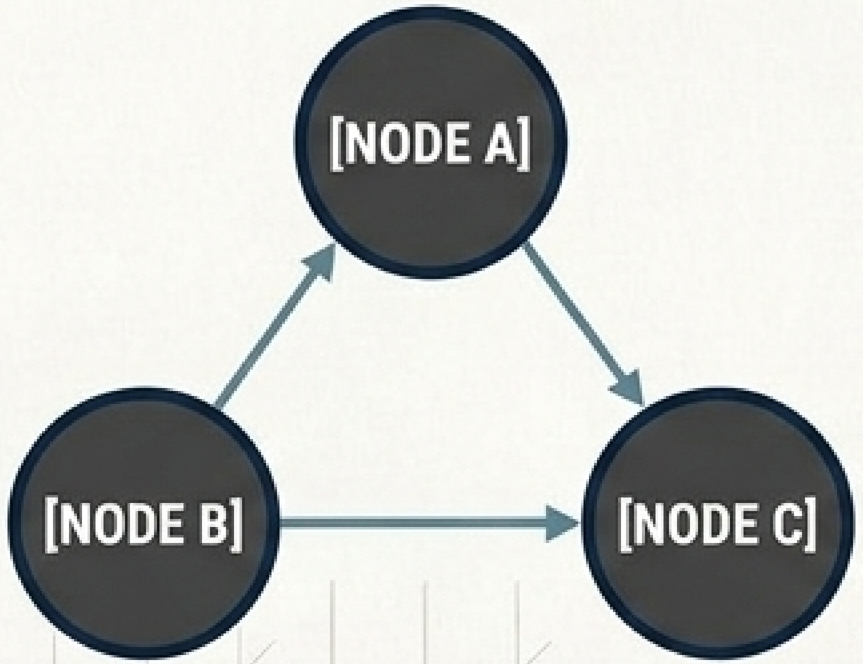
Part 3

Result: 15.2 Mbit/s

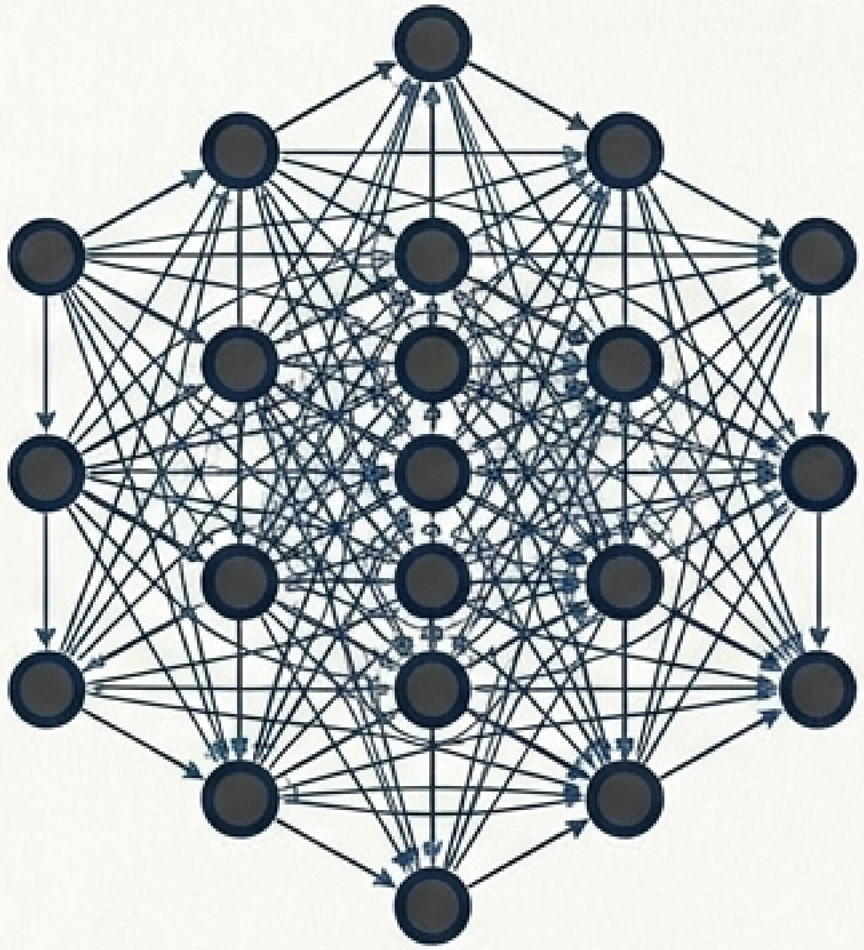
We transformed chaotic channel state changes into a precise, predictive network performance metric.

The Threat of State-Space Explosion

Simplicity

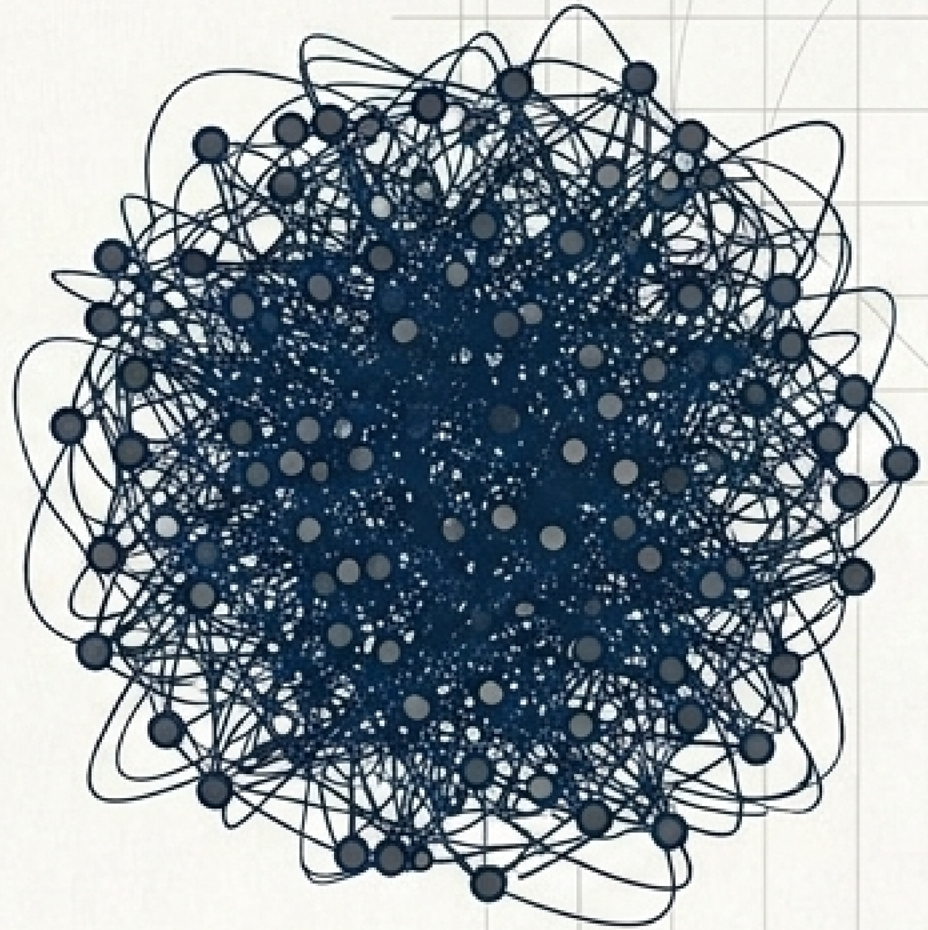


Geometric Growth



20 queue × 5 channel × 10 user
= **1,000 states**

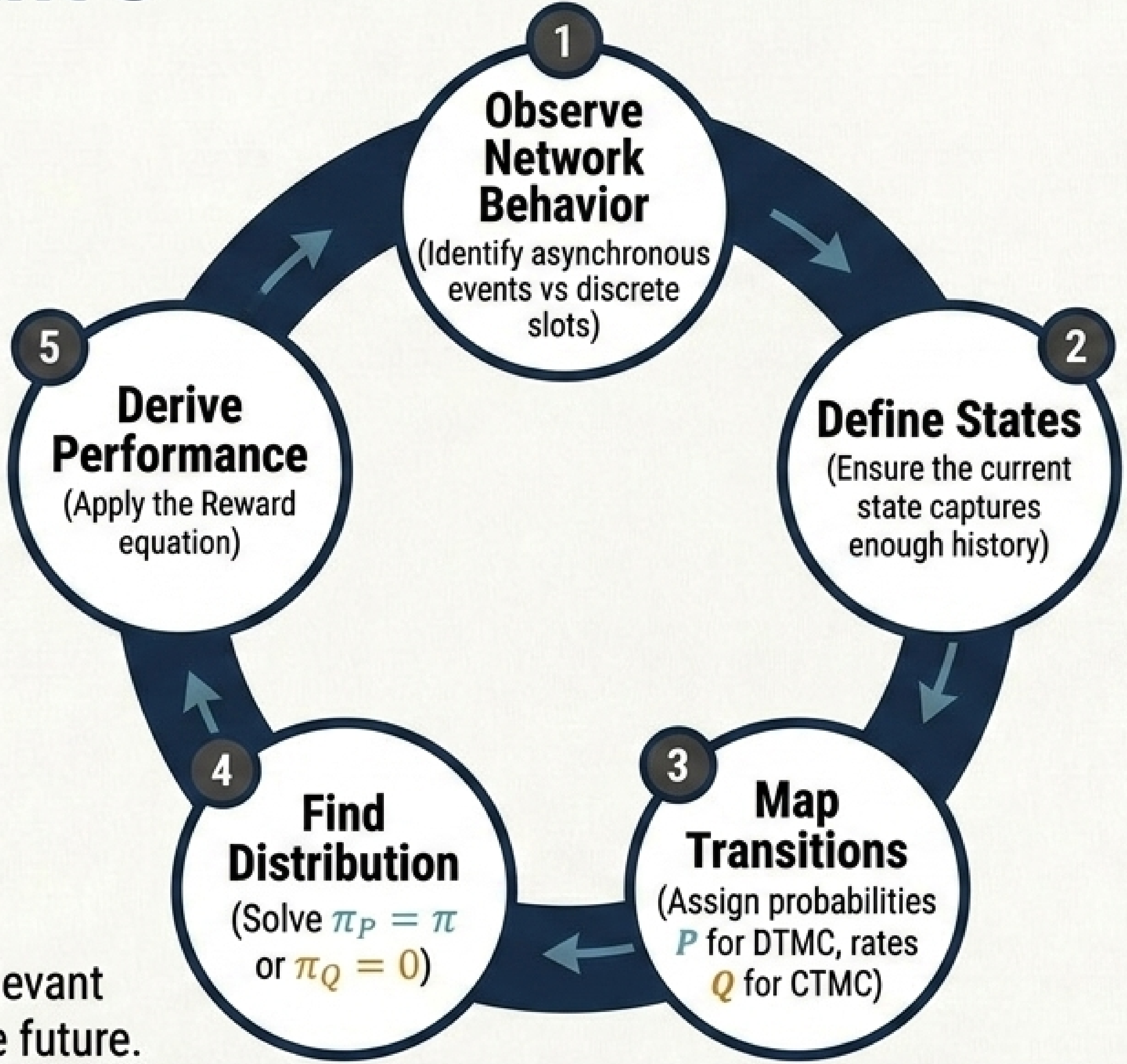
Computational Chaos



Adding just one more 20-state variable
= **20,000 states.** ($|S| = \prod n_k$)

Exact mathematical analysis becomes impossible. Solutions require state aggregation, simulation, or reduced-order models. A Markov model is an abstraction, not universally realistic.

The Consultant's Blueprint



The current state contains all relevant information needed to model the future. This is the foundation for queueing theory.

END_DIAGNOSTIC // PROCEED_TO_QUEUEING_THEORY